ROHIT SHROFF

location | number | mail | website

PROFILE SUMMARY

Highly analytical AI & Data Engineer with extensive experience architecting production-grade data pipelines, distributed systems, and enterprise GenAI solutions. Adept at transforming complex, legacy data ecosystems into predictive, automated intelligence platforms. Proven track record of leveraging Medallion architecture, LLMs, and explainable AI to drive high-scale business automation and systems assurance.

TECHNICAL SKILLS

- **Languages & Frameworks:** Python, Node.js, TypeScript, Next.js 14, React 18, PyTorch, Transformers, Express.js, Tailwind CSS.
- **AI / GenAI / MLOps:** Large Language Models (LLMs), PEFT (LoRA), PPO (Reinforcement Learning), Model Context Protocol (MCP), Explainable AI (XAI), Tokenization, Toxicity Mitigation.
- **Data Engineering & Systems:** Oracle Data Integrator (ODI), PySpark, Docker, CI/CD (Vercel).
- **Databases & Clouds:** PostgreSQL, MongoDB, Apache Cassandra, Prisma ORM, Oracle Analytics Server (OAS), OCI, AWS.

EDUCATION

BACHELOR OF ENGINEERING IN ELECTRONICS AND COMMUNICATION
BMS College Of Engineering 2020 - 2024

EXPERIENCE

Associate Consultant (AI & Data Engineer) at Company - August, 2024 - Present

Project: AI-Powered Telecom Signaling Assurance Platform

- **Architected an enterprise-grade AI Network Assurance engine** utilizing a modern Medallion (Lakehouse) Architecture to ingest and process over 10M+ raw performance, telemetry, and transaction records across 338 routing processors.
- **Engineered real-time Feature Layers (ML_*)** to calculate dynamic capacity utilization trends and rolling baselines (1h/6h/24h), shifting operations from reactive threshold alerts to proactive predictive modeling.
- **Developed an Explainable AI (XAI) Root Cause Analysis (RCA) engine** that correlates infrastructure telemetry and transaction drops to deliver deterministic operational recommendations alongside confidence scores and reason codes.

Project: Enterprise BI Core Migration & LLM Automation

- **Spearheaded a massive legacy reporting migration** from Tableau/BusinessObjects to Oracle Analytics Server (OAS), handling end-to-end data lineage translation, edge-case remediation, and schema alignment.
- **Pioneered an LLM-driven automation utility** to parse and convert complex calculation formulas from source legacy systems to OAS syntax, reducing manual developer migration hours and standardizing codebase alignment.
- **Acted as Subject Matter Expert (SME)**, driving deliverables across geographically distributed teams, performing rigorous peer code reviews, and creating a unified knowledge asset base to drastically accelerate onboarding.

TECHNICAL PROJECTS

Local MCP Gemini Automation Engine | Python, GenAI SDK, Model Context Protocol

- **Architected a local AI agent workspace** using the Model Context Protocol (MCP) and gemini-2.5-flash to execute autonomous file-system operations via secure background OS stdio text pipes.
- **Engineered an asynchronous terminal orchestration engine** that intercepts custom syntax (/format, @) to dynamically parse and inject local data structures into LLM prompt-template layers in real time.
- **Developed a decoupled tool-routing pipeline** that translates local script schemas into structured JSON tool configurations, enabling automated loops to safely read, modify, and rewrite local data blocks.
- Link

Dynamic Content Management Portfolio Engine | Next.js 14, TypeScript, Prisma, PostgreSQL

- **Architected a full-stack, type-safe portfolio application** using the Next.js 14 App Router, Next.js API Routes, and a PostgreSQL database layer hosted via Vercel.
- I**mplemented a secure, customized admin control panel** utilizing NextAuth.js for credential-based authentication and route protection middleware, facilitating real-time CRUD data management.
- **Optimized database access performance** by leveraging Prisma ORM to map structural tables, removing the need for page redeployments by achieving real-time schema rendering via Server-Side Rendering (SSR).
- Link

Dialogue Summarization Enhancement & Detoxification | Python, PyTorch, Transformers, RLHF

- **Fine-tuned a flan-t5-base LLM for dialogue summarization** using Parameter-Efficient Fine-Tuning (PEFT) and LoRA, preserving computation power while matching baseline performance metrics.
- **Implemented Reinforcement Learning from Human Feedback (RLHF)** via Proximal Policy Optimization (PPO) to systematically minimize model output toxicity levels.
- **Integrated a RoBERTa-based hate-speech validator** to dynamically audit generation metrics, achieving verifiable improvements in ROUGE scores alongside safer AI generations.
- Link

Big Data Streaming & Analytics Pipeline | Kafka, Airflow, PySpark, Cassandra, Docker

- **Architected an end-to-end distributed data pipeline** inside Docker containers to consume and analyze heavy streaming data telemetry.
- **Orchestrated an Apache Airflow DAG** to continually poll APIs, streaming live user-data payloads directly into Confluent Kafka queues.
- **Utilized Apache Spark Streaming (PySpark)** for live sliding-window computations before committing clean payloads into an Apache Cassandra NoSQL cluster.
- Link

Resumot: Local RAG Resume Intelligence Engine | Python, Llama.cpp, DeepSeek-R1 (GGUF), ChromaDB, Streamlit

- **Architected an end-to-end, privacy-centric Retrieval-Augmented Generation (RAG) platform** to parse and analyze unstructured PDF resume data using pure local inference without relying on cloud APIs.
- **Engineered a vector database pipeline utilizing SentenceTransformers (all-mpnet-base-v2)** to chunk, embed, and index multi-source textual data into a local ChromaDB vector store for semantic similarity retrieval.
- **Optimized token-constrained local inference** by leveraging llama-cpp-python to execute a quantized DeepSeek-R1 13B model in .gguf format, implementing dynamic context injection and stream-rendered query responses.
- Link

COURSES AND CERTIFICATIONS

- OCI Certified Data Science Professional – 2025 (Link)
- OCI Certified Generative AI Professional – 2025 (Link)
- Oracle AI Vector Search Certified Professional – 2025 (Link)
- Oracle Fusion AI Agent Studio Foundations Associate – 2025 (Link)
- Oracle Analytics Cloud Certified Professional – 2024 (Link)
- OCI Certified AI Foundations Associate – 2024 (Link)
- OCI Cloud Infrastructure Foundations Associate – 2024 (Link)
- Generative AI with LLMs – DeepLearning.AI & AWS (2023) (Link)
- Machine Learning Specialization – Stanford University (2024) (Link)
- Oracle Data Integrator 12c Certified Implementation Specialist (Link)
- Introduction to Model Context Protocol (Link)